Fail-closed verification of model-drafted claim-denial appeals against the clinical record: a 50-case synthetic benchmark of three Meta models

Preprint, doi:10.5281/zenodo.24000088. Describes repository version 1.2.0 (commit 75fe542, gate v1.7), deposited at doi:10.5281/zenodo.23935951.

James H. Rosing, MD, FACS

November 5, 2026

Newport Beach, California. ORCID 0009-0009-0911-1915. Correspondence: jamesrosing@gmail.com.

Keywords: claim denial; medical necessity appeal; prior authorization; large language models; fabrication; fail-closed verification; Medicare Advantage; synthetic data.

Abstract

The insurer has given the denial to a machine, and the machine answers in seconds. The physician has been given nothing, and must meet in hours what was done in an instant, so most denials are never met at all. The reply owed is a writ in the oldest sense, an instrument issued under a signature that binds its author to every word; a machine may draft it, but a draft that asserts one thing the record does not hold, over a physician's name, is worse than none. I asked whether Meta's models, one of them open weights, could prepare that instrument from the chart with every sentence held to the record by a deterministic check, decline when the record could not support one, at a cost and speed a practice could use, and whether the same check, pointed the other way, would hold the insurer's own determination to the same record. Fifty synthetic claim denials across eighteen high-dollar services, thirty answerable from the chart and twenty not, $940,000 of billed care, were generated with a grading key. Three Meta models drafted responses with the chart in the prompt, by tool calls, and with a live query of a coverage index; a gate reimplemented from the LITMUS papers verified every quote, number, date, code, and side against the cited entry, and 12,242 injected fabrications measured its sensitivity. Post hoc, blocked drafts were sent back once with the gate's reasons, two models ran at higher effort, and the denials were read back from letter images. Across nine kept runs, agreement with the key was 44 to 50 of 50 with no fabricated quote in any sentence; muse-spark-1.3 defended the full $568,500 of the warranted cases in two of its three runs at $0.63 to $0.96 a run; the gate caught 8,602 of 8,602 hard corruptions and none of 2,382 negation or pronoun changes, at 4 to 16 seconds and $0.007 to $0.054 a case; one draft passed on an unsupported case. One repair turn passed six of seven blocked drafts and eroded no decline; higher effort bought muse-spark-1.3 no agreement and the open-weights model 50 of 50 at medium; the letters were read with all eight fields correct in 46 of 50, every error one glyph. Pointed at the plan's determination, the gate refuted the one fabricated quote and passed a denial whose every sentence was in the record and whose reading was wrong: one chart, two verified conclusions, no side taken. The gate verifies that a statement is in the record, not that the record satisfies the criterion; the machine prepares the instrument and cannot issue it, and the physician's signature is what makes the draft a writ.

1 Introduction

1.1 The problem

In 2023, Medicare Advantage insurers made 49.8 million prior authorization determinations and denied 3.2 million of them, 6.4 percent. About 11.7 percent of the denials were appealed, and 81.7 percent of those appeals were overturned [1]. Derived from those rates, about 374,000 denials were appealed, about 306,000 were reversed, and about 2.8 million were never challenged at all (Figure 1). When denials are challenged, most fall. Most are never challenged, because the cost of answering lands on the people with the least time: physicians report 13 hours a week spent on prior authorization, and 79 percent say patients have abandoned treatment while waiting [2].

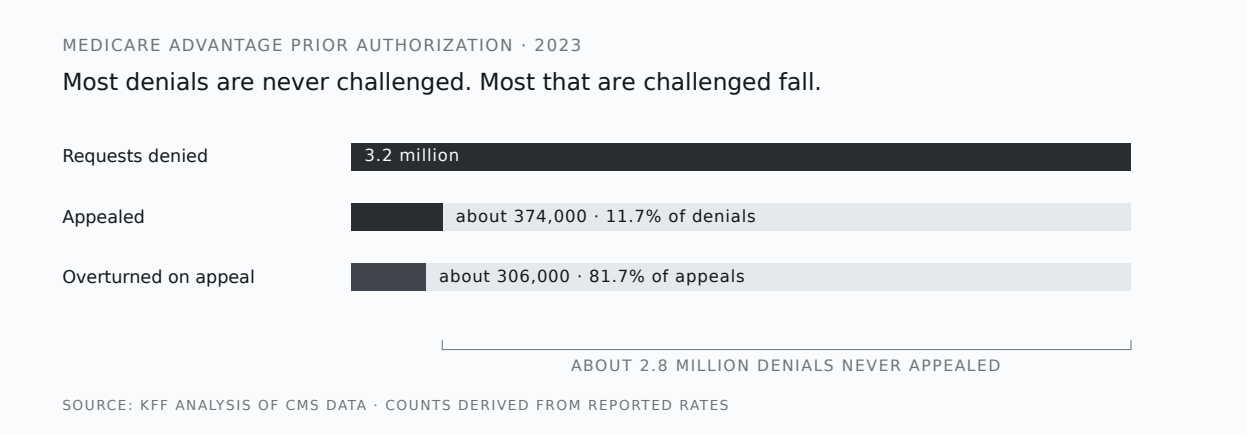


Figure 1: Medicare Advantage prior authorization, 2023: requests denied, appealed, overturned, and never appealed, derived from the reported rates [1].

The rules are not uniform. Each payer writes its own, and in 2022 the Inspector General of the Department of Health and Human Services found Medicare Advantage organizations applying clinical criteria that Medicare itself does not use, and denying 13 percent of a sample of requests that met Medicare coverage rules [3].

The surgeon whose judgment is overridden spent four years in medical school and five or more in residency and specialty training, and carries the license, the liability, and the duty to see the patient through. The decision to refuse carries none of that weight. I do not think most of the people who build these systems set out to harm patients. But a system rewarded for saying no, and rarely asked to show its reasons, will say no more often than it should.

1.2 The public record

The automation of the denial is documented in litigation, a Senate investigation, and the press (Table 1). The insurers dispute these characterizations: UnitedHealth says nH Predict only guides care planning, called the Wall Street Journal's analysis of its diagnoses inaccurate and biased, and says it has full confidence in its practices; Cigna says PxDx is not artificial intelligence.

Table 1: Table 1. Automated denial in the public record, 2019 to 2026.

Who	Tool	What is alleged or reported	Status
UnitedHealth (naviHealth)	nH Predict	Predicts post-acute length of stay for Medicare Advantage patients; plaintiffs allege about 9 in 10 denials were reversed on appeal	Class action in discovery, D. Minn.; March 2026 order compels records back to 2017 [4]
UnitedHealthcare	Post-acute review	Post-acute denial rate rose from 8.7 percent (2019) to 22.7 percent (2022)	Senate Permanent Subcommittee on Investigations report, October 2024 [5]

Who	Tool	What is alleged or reported	Status
CVS Health	Post-Acute Analytics	Project to reduce skilled nursing spending; savings target raised to $77.3 million over three years	Senate report, October 2024 [5]
Cigna	PxDx	More than 300,000 claims denied in two months of 2022, averaging 1.2 seconds each	Class action, E.D. Cal., filed July 2023 [6]
Medicare Advantage insurers	Insurer-initiated diagnoses	Diagnoses added by insurers rather than a treating physician drew about $50 billion from Medicare over 2018 to 2021; about 66,000 patients coded with diabetic cataracts after cataract surgery	Wall Street Journal series, 2024 [7]; in July 2025 UnitedHealth disclosed Department of Justice criminal and civil investigations of its Medicare billing [8]
CMS	WISeR	Model of AI-supported prior authorization in traditional Medicare across six states, for skin substitutes, nerve stimulators, knee arthroscopy for osteoarthritis, cervical fusion, and other services	Live since January 2026; repeal resolution pending [9, 10]

1.3 A standard, not a product

No machine should make a claim about a patient's care that it cannot support. It must cite its source or show its derivation. If it cannot, it stops, and the physician decides. I wrote that standard down as LITMUS and published it openly under CC BY 4.0, so anyone can use it, test it, or argue with it [11, 12]. It applies to an insurer's algorithm as much as to mine. The position paper argues that an autonomous system can be trusted only in proportion to the strength of the external check that verifies it, that medicine has one valid check, the certified specialist physician, and that the machine may act alone only where consensus is explicit and settled and must defer to the physician everywhere else. The architecture companion specifies the implied system: a model-pluggable agent runtime, an extensible registry of tools, and a fail-closed verification spine whose criteria are curated per specialty, with every claim, gate decision, and verdict recorded in a provenance layer.

1.4 What is not known

Large language models can write a medical-necessity letter that reads well. Whether one can write such a letter without stating anything the chart does not contain, under a check that can prove it did not, has not been measured on a set anyone can rerun. Nor has the second half of the task been measured: whether the model declines when the record cannot answer the denial, including the administrative denials for which a medical-necessity appeal is the wrong instrument. The LITMUS papers predict the shape of the answer, a deterministic check that verifies presence in the record and not sufficiency against the criterion, but they contain no measurement.

1.5 Objective and hypothesis

The objective was to measure, on a fixed synthetic set with a grading key, whether three Meta models, one of them open weights, can draft the physician's response to a claim denial from the chart with every sentence verified against the record, decline when the record cannot answer, and do so at a cost and latency a practice could use. The hypothesis: a deterministic referential gate reduces fabricated statements in model-drafted appeals to zero on this set, at the cost of some true derivations blocked, without reducing agreement with the key below ninety percent. The secondary question was whether a model given the chart as an index and two tools, rather than whole, reads what a reviewer would read.

2 Methods

2.1 Study design and setting

A bench study on synthetic data, run during the Meta Global AI Developer Hackathon, October 26 to November 4, 2026 [13]. No human subjects, no patient data, no institutional review. Every line of code was written during the event; a ledger has a row for every commit, and a build log is regenerated from the git history

and the result files after every push [14]. The line between what was fixed before the reported runs and what changed after them is drawn by the git history, not by memory (Table 2).

Table 2: Table 2. Pre-specified and post hoc elements, by the git history.

Fixed before the kept runs began (October 27, 01:17 PT)	Changed after, each with a ledger row
The fifty cases and their grading keys. Four keys were corrected during development runs between 00:40 and 01:17 PT, each defect found by a model reading the chart (section 4.4); no case file changed after the kept runs began.	The agentic mode and its two chart tools (October 27, 20:22 PT) and the live policy query (October 27, 23:44 PT).
The gate (v1) and the appeal contract: one citation per sentence, a verbatim quote, every hard token sourced. Fact sourcing was widened from the quoted span to the whole cited entry after a development run (00:50 PT).	Gate v1.1 to v1.7 (section 2.4). Every kept run was re-gated under each version with the change recorded in the file; re-gating to v1.3 changed seven sentence labels and no verdict, to v1.4 and v1.5 none, to v1.6 none in the kept runs and four verdicts in the repeat and payer-side runs, and to v1.7 four verdicts in the kept runs and five elsewhere, each reported in section 3.
The primary measure, agreement with the key, and the outcome categories.	The advisor, the mutation harness (October 28), the repeat runs (October 28 to 29), the payer-side runs (October 29), the viewer, and this document.
Three models, effort low, eight cases in flight.	The repair turn and the stated reason, the effort sweep, the letters and their intake, and the dictation pipeline (October 30). Every run from these is post hoc, sits in its own directory, and leaves Table 3 unchanged.

2.2 The system under test

A denial arrives with its reason code and the payer's remark (Figure 2). The model is given the payer's written criteria, the denial, the case facts, and the chart, and asked for one of two things: a decision that the record does not support an appeal, with the reason, or a draft in which every sentence carries one citation to one chart entry and a verbatim quote from it. The gate then checks every sentence. If the record does not support an appeal, the software says so instead of arguing anyway. Nothing is sent without a physician's signature. The viewer renders a passing draft as that instrument: the payer's reason quoted at the head, each sentence footnoted to the entry, date, and verbatim quote the gate verified it against, the gate version and the input hash at the foot, and a blank signature line under a band that reads draft, unsigned, not sent. It can be printed or saved as text; the page sends nothing. The software gives no advice to patients, recommends no treatment, and sends nothing on its own.

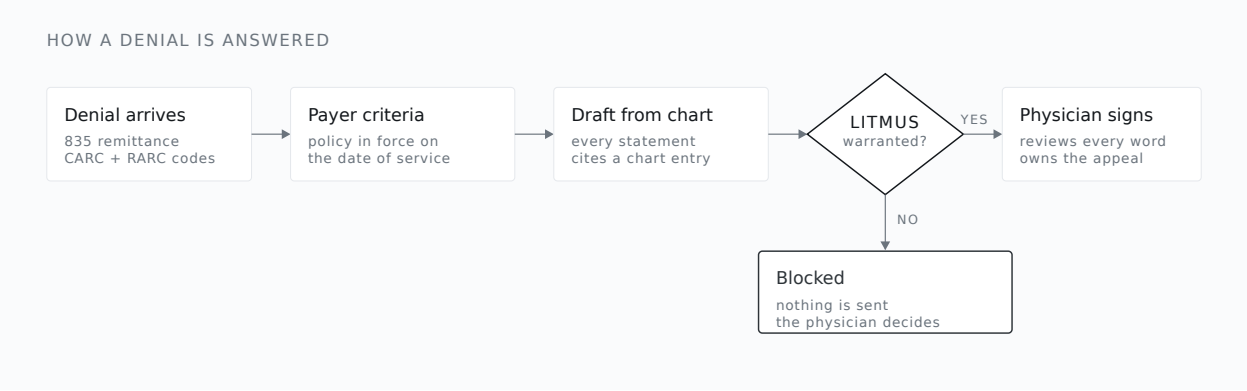


Figure 2: The system under test. A denial arrives as a remittance with its reason code; it is read against the payer's criteria in force on the date of service; a response is drafted from the chart with every sentence citing a chart entry; the gate checks every sentence; a passing draft goes to the physician to read and sign, and a failing draft is blocked with the reason.

2.3 The fifty cases

What kind of denial. Every case is a claim denied after the service was given: the payer's remittance, carrying a claim adjustment reason code (CARC) and the payer's remark, for a procedure, admission, drug, or continued stay already delivered. None is a prior-authorization request refused before care. The statistics in section 1.1 are prior-authorization figures because that is where the public record is richest; the response to either kind of denial is the same document, the physician's record read against the payer's written criteria, and the set should be read as a test of that document, not of the prior-authorization process. Thirty-four cases carry CARC 50, not medically necessary. Three carry CARC 197, no authorization on file: in two the plan allows a retrospective medical-necessity review and the chart answers it, and in one the plan refuses that review, so a medical-necessity appeal is the wrong instrument and the right answer is to decline. The rest are experimental or investigational (CARC 56), non-covered (96, 167), diagnosis or modifier inconsistent with the procedure (11, 4), frequency (151), benefit maximum (119), and timely filing (29).

How they were made. The set is generated, not hand-written. A generator builds all fifty deterministically from a fixed seed (20261027) using 18 service templates, one per service, and a test checks that the committed files match the generator, so anyone can regenerate them and get the same bytes. A fixed plan allocates 30 warranted cases, where the chart documents every criterion, and 20 that are not, in four kinds: 6 where the chart never addresses a criterion, 7 where it raises the criterion but falls short, 4 administrative denials no chart can answer, and 3 plan exclusions. One case, case-023, was relabeled from the first kind to the second on October 30, after the archived version, with the reason in section 4.6; its text is unchanged. Each template writes the chart in dictation voice, the way a clinician speaks a note rather than the way a payer checklist reads, and the generator adds the routine traffic a real chart has, a referral letter, a blood-pressure check, a pharmacy call, so that about two thirds of the 19 to 35 entries (mean 25.6) are unrelated and a citation has to land on the right one. The six payers are fictional, alternating a standard and a strict criteria profile, and the criteria attributed to them were written for this benchmark from publicly described policy patterns; they are not any insurer's policy. A patient is an alias, an age, and a sex. Each case carries its grading key: whether an appeal is warranted, which criteria the chart meets, which entries support each, which entries are unrelated, and a one-line rationale. The key was written by the same person who wrote the templates, and four of its labels were wrong until the models found them (section 4.4).

Read as an experiment, the thirty warranted cases are positive controls, the twenty others are negative controls of four kinds, the unrelated entries in every chart are the noise floor for citation precision, and the injected fabrications of section 2.7 are spike-ins.

2.4 The gate

The gate was reimplemented during the event from the published papers and not copied from any prior system. It is deterministic and sits outside the model. For each sentence it resolves the citation to one chart entry; requires the quote to appear verbatim in that entry, with whitespace, case, quote glyphs, dash variants, and compatibility forms such as fullwidth digits normalized and nothing else, and at least one word of it not a stopword; and requires every hard token in the sentence to appear in the cited entry, the payer criteria, the denial, or the stated case facts. A hard token is a number (written in digits, in words, as an ordinal, as the article a or an before a unit, or as a Roman numeral after a grading word), a date (in any written form, held as one), a code (letter, system word, and digits), or a side (left, right, or an abbreviation of either). The day and the month of a date are not admissible as bare numbers. From v1.7 a number set beside a side keeps that side: where the sources pair the number with the other side, the sentence is refuted, so "690 on the right" is not sourced by an entry that puts 690 on the left. A sentence that fails is refuted with the token named, or, from v1.3, returned as needs-citation when every missing token lives in some other entry of the same chart. A check that throws is an error, never a pass. A draft passes only when every sentence verifies and every criterion the draft claims has verified evidence; otherwise it is blocked. The gate has no resemblance threshold and no keyword list. Its version is stamped into every verdict, and because it is a pure function of its inputs, any stored draft can be re-gated under a later version and the change recorded.

The versions after v1 each added a token class or closed a defect, with a test: compound number words read as one number and hundreds included (v1.1, v1.2); the needs-citation label (v1.3); a side treated as a hard

token wherever it names one (v1.4); a side word inside an idiom admitted from no source, in the entry as well as in the sentence (v1.5); the six classes an external review found (v1.6, section 4.4); and the article, ordinal, Roman numeral, and side-number pairing classes (v1.7).

The same gate runs in the viewer. Once a replay has reached its verdict, any sentence can be edited in the page, its words, its quote, or its citation, and re-checked by the gate bundled into the browser from the same module and the same admissible sources; the result is stamped with its reason, nothing is saved, and the stored verdict does not change. It is the spike-in of section 2.7 as a control a reader can turn by hand.

2.5 Models, modes, and the live policy query

Three models were tested: muse-spark-1.3 and muse-wonder-1.0 through the Meta Model API, and Muse Spark Open, the open-weights model, hosted on Baseten, all at effort low with eight cases in flight (three for Baseten after a run at eight lost cases to rate limiting). The three kept Baseten runs of October 27 and 28 used the model id meta-models/muse-spark-open on the workspace key, and every later run the hackathon id meta-models/muse-spark-open-1026; both serve the same model, and each result file records which. Each model ran in three modes. In the first the whole chart is in the prompt. In the second, the agentic mode, the model receives an index, one line per entry with the id, date, kind, and opening words, and two tools, list_entries and read_entry, and reads what it decides it needs; every tool call is kept in a trace. The third adds a tool, policy_status, which queries CHART, the author's pre-existing coverage index of United States payer policy, for when the named authority's coverage material last changed and was last checked. It returns no policy text, so the criteria in the request remain the criteria, and its answer is labeled as a live query against a prior-work data source wherever it appears. The viewer replays any case from any kept run one tool call at a time [15].

2.6 The advisor

After the gate, a second model (muse-wonder-1.0) is shown each criterion with the quotes the draft cited for it and asked one question: do these quotes, read literally, satisfy the criterion? It answers supports, partial, or silent, with one sentence why. The answer is stored with the model's name and shown beside the criterion in the viewer. It never feeds the gate and never changes an outcome; it is an opinion with a name on it, for the physician to weigh.

2.7 Outcome measures and analysis

The primary measure is agreement with the grading key: a case counts as agreed when the model drafts an appeal for a case the key labels warranted and the gate verifies every sentence, or when the model declines a case the key labels unsupported. The other outcomes are blocked (the model drafted and the gate stopped it), missed (a decline on a warranted case), false pass (the gate passed a draft on a case the key labels unsupported), and failed (no valid answer); the five sum to 50. Secondary measures: fabricated quotes (a quote not found verbatim in the cited entry, in a sentence or in an evidence item, on any draft including declined ones), sentences verified, citation precision (the share of cited entries the key names as evidence), entries read per case and the share of the key's evidence entries opened (agentic modes), care defended (the billed value of the warranted cases whose appeal passed, out of $568,500 across the thirty warranted cases and $940,000 across all fifty, reference values of the synthetic set and not a payer's), median latency, and cost per case from the provider's token prices.

Spike-in. The gate's sensitivity cannot be read off runs in which no model fabricated anything, so it was measured by spike-in: each sentence the kept runs verified was corrupted one way at a time and re-gated. Corruptions were classed as hard (a number replaced by a value absent from every admissible source, the citation moved to an entry lacking the quote, a word dropped from the quote, a date shifted, a side swapped, and the classes added at v1.6 and v1.7), collision (a wrong number or side that happens to appear in an admissible source), and soft (a negation flipped, a pronoun swapped). Sensitivity is the share caught, where caught means the gate returned refuted or needs-citation; where none was missed, the 95 percent upper bound on the miss rate is given by the rule of three. Agreement counts carry 95 percent Wilson intervals.

Run-to-run variance was measured by repeating each live-query configuration with the same cases and settings. Fifty cases is a small set: one case is two percentage points.

The other chair. A post hoc run put the gate on the other side of the exchange, added October 29 after the kept runs. The same three models, over the same fifty charts and criteria, were asked for the plan's written claim determination instead of the physician's appeal, under the same contract and the same gate, with the key inverted: a denial is warranted exactly where the record fails a criterion. The outcomes mirror the primary ones: declined to deny where the record supports the service, a verified determination where it does not, a determination written against a record that supports the service (the symmetric false pass), blocked, and failed. These runs are kept apart from Tables 3 and 4 and from the viewer's scoreboard.

2.8 Post hoc additions of October 30

The repair turn. The harness took a repair option (default 0; every kept run was at 0). When the gate blocks a warranted draft and a repair remains, the harness sends the model one more user turn in the same exchange: the gate's record, verbatim, one line per finding, each unverified sentence with its index, status, and reason, and, for a needs-citation, the entries the gate names as holding the token, then the draft-level reasons. The model is asked for the complete revised draft under the same schema, keeps its tools and its earlier reads in the agentic mode, and is told that declining is an answer. The revised draft is re-gated under the same gate. A declined draft is never offered a repair: the model's decision to decline stands. The result file keeps the first gate record and the outcome scored on it beside the final outcome, and every repair turn with the reasons sent, the revised draft, its gate record, its tokens, and its latency. The summary reports separately how many repairs turned a blocked draft on an unsupported case into a pass, which would be the repair eroding the decline. The three live-query configurations were run again with one repair.

The stated reason. The Meta Model API returns the model's reasoning only as redacted thinking blocks, so there is no thought text to show. Instead, the two tools that open something, read_entry and policy_status, took a required field, why: one sentence on what the model expects the entry to bear on and why it is opening it now. The executor ignores the field; the trace records it, and the viewer shows it under the chart column as the entry opens. The repair runs are the first to carry it.

The effort sweep. muse-spark-1.3 and Muse Spark Open were run again in the live-query configuration at effort medium and at effort high: the same 50 cases, the same tools, the same 12,288-token output cap, the same gate. Effort is the API's output_config.effort, set explicitly on every call; on the Meta Model API the reasoning it buys is reported as thinking tokens and billed as output, and on the Baseten endpoint it is not reported. muse-wonder-1.0 was not swept: its kept run agreed on 50 of 50 and it is the costliest of the three.

The denial as a letter, and its intake. A practice does not receive a case file; it receives a remittance. Each of the fifty denials was rendered as a one-page remittance letter image, 1275 by 1650 pixels, deterministic from the case file: a fictional payer letterhead, the patient alias with age and sex, one claim line with the code and its code system, the dates of service, the billed amount and nothing allowed, the CARC and its text, the remittance advice remark code (RARC) where there is one, the payer's remark, and an appeal-rights paragraph with that payer's appeal window. No member, subscriber, or claim identifier appears, because the set issues none, and the page says so. Each image was sent to muse-spark-1.3 as an image block with a schema for eight fields (the CARC, the remark, the first and last dates of service, the code, the code system, the billed amount, and the payer's name), and each field read was scored against the case file: exact after trimming for codes and dates, exact after whitespace and quote normalization for the remark and the payer's name, a number for the amount, empty for no service end. This is document intake of an administrative letter and nothing clinical: no chart entry is sent, no clinical judgment is asked for, and the model is told to transcribe what is printed and to leave a field empty rather than guess.

Dictation. The chart entry a draft quotes is, in practice, dictated. A command transcribes a recording with Muse Voice Transcribe through the Meta Model API, writes the transcript as that entry's text into a copy of the case with the original kept beside it, and runs the primary configuration on that one case, so that every quote the gate verifies against the entry is a quote of speech, and a transcription error becomes a fact of the record the gate will hold the draft to. The pipeline was proven end to end on a test clip of the same entry read by a text-to-speech voice (28.8 seconds; not the author's voice, and not committed); the author's own

recording is not yet made, and the viewer does not show the stand-in.

2.9 Reproducibility

The test suite holds 282 tests. The fifty cases regenerate byte for byte from the seed. A harness runs a model over the set and writes a result file whose header records the model, mode, gate version, and any later re-gate. The spike-in reruns from the stored result files. The build log and the tables of section 3 regenerate from the result files; nothing in those tables is typed by hand. The gate ships on its own as a command-line tool and a Model Context Protocol server, published on npm as litmus-gate [16], so a drafting agent on any host can be held to the record before its draft is shown to anyone; a test checks that the package reproduces every stored verdict and hash of a kept run. The viewer replays any case from any kept run, lays the three models' results for one case side by side, and is checked in headless Chromium. The repository is deposited at Zenodo with a DOI for each version [14].

3 Results

Tables 3 to 9 are generated from the result files. Spark 1.3 is muse-spark-1.3 and Wonder 1.0 is muse-wonder-1.0, both through the Meta Model API; Spark Open is Muse Spark Open, the open-weights model, on Baseten. All runs are at effort low unless the table says otherwise. All verdicts are under gate v1.7.

Table 3: Table 3. Kept runs: nine runs of 50 cases, headline outcomes. Agreed is correct pass plus correct decline, with its 95 percent Wilson interval. Care defended is the billed value of the warranted cases whose appeal passed, of $568,500. Cost is per case.

Model	Mode	Agreed	95% interval	False pass	Fabricated quotes	Care defended	Cost
Spark 1.3	chart in prompt	50 of 50	93 to 100%	0	0	$568,500	$0.013
Wonder 1.0	chart in prompt	49 of 50	90 to 100%	0	0	$560,500	$0.037
Spark Open	chart in prompt	49 of 50	90 to 100%	0	1	$548,500	$0.007
Spark 1.3	agentic	49 of 50	90 to 100%	0	1	$543,500	$0.017
Wonder 1.0	agentic	49 of 50	90 to 100%	0	0	$568,500	$0.052
Spark Open	agentic	48 of 50	87 to 99%	0	0	$503,500	$0.018
Spark 1.3	agentic, live query	49 of 50	90 to 100%	0	0	$568,500	$0.019
Wonder 1.0	agentic, live query	49 of 50	90 to 100%	0	0	$543,500	$0.054
Spark Open	agentic, live query	44 of 50	76 to 94%	1	0	$445,500	$0.016

Table 4: Table 4. Kept runs, secondary measures of the same nine runs. Entries read is per case of a mean 25.6 (agentic modes); latency is the median per case.

Model	Mode	Blocked	Missed	Failed	Sentences verified	Citation precision	Entries read	Latency
Spark 1.3	chart in prompt	0	0	0	247	98%	all	16.3 s
Wonder 1.0	chart in prompt	0	1	0	236	97%	all	8.8 s
Spark Open	chart in prompt	1	0	0	233	95%	all	4.3 s
Spark 1.3	agentic	1	0	0	264	95%	9.6	8.0 s
Wonder 1.0	agentic	1	0	0	244	96%	8.7	10.0 s
Spark Open	agentic	1	0	1	239	95%	9.4	9.9 s
Spark 1.3	agentic, live query	1	0	0	263	94%	9.8	10.8 s
Wonder 1.0	agentic, live query	1	0	0	228	94%	7.9	9.0 s
Spark Open	agentic, live query	5	0	0	244	97%	9.3	7.3 s

Agreement. Across the nine kept runs, agreement with the key was 44 to 50 of 50 (Table 3). muse-spark-1.3 agreed on 50, 49, and 49; muse-wonder-1.0 on 49, 49, and 49; Muse Spark Open on 49, 48, and 44. The full $568,500 was defended in 3 of 9 runs, at $0.63 to $2.61 for the fifty cases of a run. In the repeat runs (Table 5), muse-wonder-1.0 with the live query agreed on 49 of 50 in all three runs, muse-spark-1.3 ranged from 47 to 50, and Muse Spark Open from 42 to 48 across five runs on two keys and two model ids.

Table 5: Table 5. Repeat runs: each live-query configuration run again with the same 50 cases and settings (8 files); ranges include the kept run.

Configuration	Runs	Agreed, range	Blocked, range	Missed	False passes	Failed	Fabricated quotes
Spark 1.3, agentic, live query	3	47 to 50 of 50	0 to 3	0	0	0	0
Wonder 1.0, agentic, live query	3	49 of 50 in all	1 in all	0	0	0	0
Spark Open, agentic, live query	5	42 to 48 of 50	2 to 7	0	2	2	0

Fabrication. No sentence in any kept or repeat run carried a fabricated quote: across 17 runs, every quote in every sentence the models wrote was found verbatim in the entry it cited, and so were the quotes of the 2,198 sentences the kept nine verify under v1.7. Two fabricated quotes sit in evidence items rather than sentences, both on drafts the model itself declined: Muse Spark Open with the chart in the prompt attached to case-040 an infliximab dosing line that the cited note does not contain, and muse-spark-1.3 by tool calls attached "three days of therapy" to case-023 against a note that does not say it. The gate refuted both items and both declines stood. One further fabricated quote occurred in a superseded run before the kept ones: muse-wonder-1.0 wrote "Proceededed" inside a quotation and the gate refuted the sentence.

What the gate blocked. The kept runs contain eleven blocked drafts under v1.7, eight when they were made: nine on cases the key labels warranted (a true derivation the gate could not source; an appeal not sent) and two on cases it labels unsupported (a warranted answer with nothing to verify, stopped). The two were answers that set warranted to true with no evidence and no sentences on cases the key labels absent (muse-wonder-1.0 agentic; muse-spark-1.3 with the live query). The rest were Muse Spark Open sentences that were true to the chart but cited one entry for a fact that lives in another: a count of three antianginal agents cited to a note that lists them without the number (three runs), a 150-foot goal from the plan of care cited against a discharge note reporting 160 feet, "bilateral" from a different imaging line, and "8 weeks of PT" cited to an injection note; and the three drafts v1.7 added, true sentences that state a count the entry gives in other words: "three visits in a week" against "three physician visits this week" (case-024), "well over a year" against "15 months" (case-006), and "second-floor" and "stage IV" where the figure lives in an entry the sentence did not cite (case-001, case-008). In the repeats, muse-spark-1.3 once answered three absent cases warranted with no evidence and the gate blocked all three. Muse Spark Open returned an answer that was not valid JSON once in the kept runs and twice in the repeats, each recorded as failed.

The false pass. One draft passed the gate on a case the key labels unsupported: Muse Spark Open with the live query, case-023, a rehabilitation admission whose chart never documents the intensity criterion of three hours of therapy a day. The draft cited "plan 5 to 6 days a week" for that criterion; the words are in the cited entry, and the gate passed the sentence. The same false pass recurred in one of the four Muse Spark Open repeats. The advisor, shown the quote and the criterion, wrote that the quote "addresses a plan for 5 to 6 days per week, but is missing the at least 3 hours of therapy per day and evidence that the patient could reasonably be expected to actively participate."

Reading behavior. In the agentic modes the models read between 7.9 and 9.8 of 25.6 entries per case and reached 93 to 98 percent of the entries the key names as evidence. A modifier denial (CARC 4) was declined without opening a single entry. Cases labeled absent were read almost completely before the model declined; one chart of 27 entries was read entry by entry. Citation precision in the warranted drafts was 94 to 98 percent.

The live query. All three models called policy_status in all 50 cases: muse-spark-1.3 in 41 as its first move, muse-wonder-1.0 in all 50 as its first move, Muse Spark Open in 40. None wrote any of its answer into a draft.

The advisor. muse-wonder-1.0 advised the three live-query runs, 91 drafted cases and 363 criteria, for $2.47. It called 73 percent of the cited quotes sufficient and 27 percent partial, never silent, and flagged the false pass on exactly the intensity criterion. Of its 97 partial calls, 90 were on criteria the gate had verified in drafts on cases the key labels warranted. Two of its answers were not valid JSON and are recorded as errors.

Table 6: Table 6. Gate sensitivity by corruption class, gate v1.7: 12,242 fabrications injected one at a time into the sentences the kept runs verified. Caught means the gate returned refuted or needs-citation; where none was missed, the 95 percent upper bound on the miss rate is by the rule of three.

Class	Corruption	Injected	Caught	Sensitivity	Miss rate, 95% upper bound
hard	number replaced by a value absent from every admissible source	1,204	1,204	100%	0.25%
hard	count rewritten as the word one or zero	749	749	100%	0.40%
hard	wrong number written in fullwidth digits	979	979	100%	0.31%
hard	number rewritten as an ordinal word of another value	512	512	100%	0.59%
hard	citation moved to an entry that lacks the quote	2,198	2,198	100%	0.14%
hard	a word dropped from the quote	2,145	2,145	100%	0.14%
hard	side swapped, other side absent from the cited entry	225	225	100%	1.33%
hard	side swapped and written as R or L	225	225	100%	1.33%
hard	number after a grading word rewritten as a Roman numeral of another value	51	51	100%	5.88%
hard	the letter of a code changed	31	31	100%	9.68%
hard	side swapped beside a number the sources pair with the other side	60	60	100%	5.00%
hard	count before a unit rewritten as the article a or an	132	132	100%	2.27%
hard	side added that the cited entry carries only inside an idiom	43	43	100%	6.98%
hard	date shifted	13	13	100%	23.08%
hard	date rewritten with a month name, the month moved	13	13	100%	23.08%
hard	the system word before a code swapped (CPT, HCPCS)	22	22	100%	13.64%
hard, all		**8,602**	**8,602**	**100%**	**0.03%**
collision	number replaced by a value that appears in the criteria, the denial, or the case facts	1,204	7	1%	
collision	side swapped, other side present somewhere in the cited entry	54	0	0%	
collision, all		**1,258**	**7**	**1%**	
soft	negation flipped	1,861	0	0%	
soft	pronoun swapped	521	0	0%	
soft, all		**2,382**	**0**	**0%**	

Gate sensitivity. Of 12,242 injected corruptions (Table 6), the gate caught 8,602 of 8,602 hard corruptions, a miss rate below 0.04 percent at 95 percent confidence. It passed 1,251 of 1,258 collisions (the seven caught are replacements that fused with an adjacent figure or code, "2500" and "CPT 50", and so became tokens absent everywhere, not readings of the collision) and 2,382 of 2,382 soft corruptions. Of the 2,208 sentences in the kept runs, the gate verifies 2,198 under v1.7 and returns 10 as needs-citation or refuted, all ten true to the chart. Ten of the twenty classes were added after the first spike-in and measure corruption shapes the gate had passed before: under v1.3 it had passed 48 swapped sides, every one in a context outside its anatomy word list; under v1.4 it had passed all 43 idiom leaks; under v1.5 it had passed every instance of the six classes an external review found (section 4.4), 2,021 of them when injected; and through v1.6 it had passed the 755 instances of the article, ordinal, Roman numeral, and side-number pairing classes, the last of which had been counted among the collisions. Each earlier report is archived beside the current one.

Gate defects found by the runs, the spike-in, and review. Two defects were found by drafts the gate wrongly refuted, each fixed with a test and the run repeated: "seventy to ninety-nine percent" and "eighty-one percent" read as two numbers each (v1.1), and "one hundred ten feet" read as a bare 100 (v1.2). The 48 swapped sides were found by the spike-in and fixed as v1.4; no draft in any run had been affected. A fourth was found in review on October 31: the gate masked the idioms "left the clinic" and "right back to baseline" in the sentence but not in the cited entry, so an entry that carried the idiom made the bare word an admissible side, and a fabricated right-sided claim citing case-006 came back verified; it was fixed as v1.5 with the exact reproduction as a test, and no draft in any run had been affected. Six more were found on November 2 by Meta's coding agent reviewing the repository (section 4.4), each with a running reproduction: the number words one and zero were not numbers, so "one week of therapy" verified against an entry that says twelve; a month name was not a date, so September 14 verified against an entry dated August 14, and the day and

month of an entry's date were admissible as bare numbers, so the same entry sourced "14 weeks"; R and L were not sides; only the digits of J9271 were compared, so K9271 verified, and the system word before a code was never read, so HCPCS 19318 verified against a CPT denial; "2,200 grams" was read as 2 and 200; "of the" satisfied the two-word quote bar; and an empty sentence verified in the viewer's editor. All were fixed as v1.6 with the reproductions as tests. Re-gating the kept runs changed no verdict at any version through v1.6; under v1.6 four verdicts moved in the repeat and payer-side runs, each a true sentence that wrote a count as "one" where the entry states it in other words ("within the hour", "next week", "this week", "three disciplines"), now blocked as needs-citation naming the entry that carries a 1, the cost of reading "one" as a number, and one sentence in a kept run moved from needs-citation to verified because a bare 1 is no longer demanded of it. The first regate under v1.7 taught two masks, each now a test: "centimeters per second" had read as the ordinal two, and "three hours a day" had donated a count of one. Re-gating under v1.7 then moved nine verdicts, four in the kept runs and five in the effort, payer-side, repair, and repeat runs: eight are true sentences now blocked as needs-citation for a count the entry states in other words, and one moved the other way, in the payer-side run, where "one week before" is now verified because the entry's "first infusion" carries a 1, a collision, not a reading. Every moved sentence is in the result files with its reason.

Cost and latency. Median latency per case was 4.3 to 16.3 seconds and cost $0.007 to $0.054 (Tables 3 and 4). Muse Spark Open reported no reasoning tokens through the Baseten endpoint, so its cost may be understated if reasoning is billed as output; latency is wall-clock and unaffected.

Table 7: Table 7. The other chair (post hoc): the same models asked for the plan's written claim determination over the same 50 charts, under the same contract and gate, with the key inverted (3 files, chart in prompt, effort low). Declined to deny is a supporting record the model declined to deny, of 30; verified denial is a failing record with a determination the gate verified, of 20; false denial is a determination written against a supporting record.

Model	Declined to deny	Verified denial	False denial	Blocked	Failed	Fabricated quotes	Sentences verified	Latency	Cost
Spark 1.3	29 of 30	20 of 20	1	0	0	0	122	10.7 s	$0.013
Wonder 1.0	30 of 30	20 of 20	0	0	0	0	102	6.5 s	$0.036
Spark Open	30 of 30	17 of 20	0	3	0	1	107	3.4 s	$0.007

The other chair. Asked for the plan's determination (Table 7), muse-wonder-1.0 declined to deny all 30 records that support the service and wrote a verified determination for all 20 that do not. muse-spark-1.3 declined to deny 29 of 30 and wrote one determination against a supporting record (case-036, bariatric surgery): six verified sentences, the fourth reading "supervised weight management visit 6 of 6" as not documenting six consecutive months; the gate passed it. Muse Spark Open declined to deny all 30, wrote 18 verified determinations, and was blocked twice, both on plan-exclusion cases: once for a sentence citing one entry for a figure (the exon 19 deletion) that lives in another, and once for a quote that does not appear in the cited entry, plan-document language the model was never given ("regardless of BMI or comorbidities"), the only fabricated quote in a sentence in any full run on either side. Every determination written where the record fails named the unmet criterion against what the record does document. In a six-case probe before these runs, muse-spark-1.3 set the denial flag on case-001, a record that meets every criterion, and then wrote six verified sentences each of which supports the care; the gate passed it. The flag did not recur in the full run.

Table 8: Table 8. The repair turn (post hoc): the three live-query configurations run again with one repair, so that a draft the gate blocked on a warranted answer went back to the model once with the gate's record verbatim and the complete revised draft was re-gated (3 files, agentic, live query, effort low). A declined draft was never offered a repair. False passes created is the subset of repaired drafts on cases the key labels unsupported.

Model	Blocked, first turn	Blocked, after repair	Repaired to pass	Repaired to decline	False passes created	Agreed, first turn	Agreed, after repair	Cost, first turn	Cost, with repair
Spark 1.3	3	1	2	0	0	47 of 50	49 of 50	$0.020	$0.021

Model	Blocked, first turn	Blocked, after repair	Repaired to pass	Repaired to decline	False passes created	Agreed, first turn	Agreed, after repair	Cost, first turn	Cost, with repair
Wonder 1.0	0	0	0	0	0	50 of 50	50 of 50	$0.059	$0.059
Spark Open	4	1	4	0	0	45 of 50	48 of 50	$0.017	$0.018

The repair turn. With one repair turn offered on blocked drafts (Table 8), muse-wonder-1.0 blocked no draft on the first turn, made no repair, and agreed on 50 of 50. muse-spark-1.3 blocked 3 drafts on the first turn. Two were repaired to pass: case-007, where the first draft had one sentence and four of five criteria without verified evidence, and the repair read seven more entries and returned ten verified sentences; and case-038, where the count of three antianginal agents was returned as needs-citation naming the entry that holds it, and the repair cited it without reading anything further. The third, case-023, an empty warranted answer on a case the key labels near-miss, was returned by the repair, after nine more reads, as another empty warranted answer, and stayed blocked. Muse Spark Open blocked 4 drafts on the first turn, every one a needs-citation on a warranted case, and all 4 were repaired to pass; agreement moved from 45 to 49 of 50 when the run was made, and stands at 45 to 48 under v1.7, which blocks a fifth draft, case-006, that passed on the first turn and was never offered a repair, because the run is stored. Its one false pass, case-023, was passed on the first turn and was not created by a repair. No repair in any run turned a blocked draft on an unsupported case into a pass. The repair turns took 4.8 to 19.9 seconds for muse-spark-1.3 and 3.8 to 8.4 seconds for Muse Spark Open, and cost $0.049 of the $1.06 run and $0.038 of the $0.91 run. Every tool call in the three runs carried a stated reason, 1,402 of 1,402.

Table 9: Table 9. The effort sweep (post hoc): muse-spark-1.3 and Muse Spark Open in the live-query configuration at effort medium and high (4 files), with each model's kept effort-low run as its first row. Empty answers are drafts that set warranted to true with no sentences, which the gate blocks; thinking tokens are per case, with a dash where the Baseten endpoint reports none.

Model	Effort	Agreed	Blocked	Empty answers	False pass	Missed	Failed	Thinking tokens	Latency	Cost
Spark 1.3	low	49 of 50	1	1	0	0	0	1,608	10.8 s	$0.019
Spark 1.3	medium	47 of 50	2	1	0	1	0	3,029	14.9 s	$0.032
Spark 1.3	high	45 of 50	5	4	0	0	0	3,567	18.2 s	$0.035
Spark Open	low	44 of 50	5	0	1	0	0	none reported	7.3 s	$0.016
Spark Open	medium	50 of 50	0	0	0	0	0	none reported	13.9 s	$0.024
Spark Open	high	47 of 50	0	0	1	0	2	none reported	25.8 s	$0.043

The effort sweep. At effort medium and high (Table 9), muse-spark-1.3 agreed on 47 of 50 at medium: one draft was blocked as an empty warranted answer (case-034), one warranted case was declined with no stated reason after 7 of 27 entries (case-011), and the false pass on case-023 recurred. At high it agreed on 45 of 50 and blocked 5 drafts: four empty warranted answers, three of them on cases the key labels unsupported, and one draft with criteria without verified evidence. Its thinking tokens per case were 1,608 at low, 3,029 at medium, and 3,567 at high. Muse Spark Open agreed on 50 of 50 at medium, with no draft blocked and no false pass, and on 47 of 50 at high, where two answers stopped at the output cap before the JSON closed and were recorded as failed, and the false pass on case-023 recurred; its output tokens per case were 3,800 at medium and 7,920 at high.

Table 10: Table 10. Document intake (post hoc): each of the 50 denials rendered as a one-page remittance letter image and read back by muse-spark-1.3 into eight fields, each scored against the case file. Cost $0.0054 per letter, median latency 2.8 seconds.

Field	Correct of 50	Errors read
CARC	50 of 50	none

Field	Correct of 50	Errors read
remark	50 of 50	none
date of service	50 of 50	none
service end	50 of 50	none
code	46 of 50	the HIPPS code KDOB1 read as KD0B1 in each of the four letters that carry it
code system	50 of 50	none
billed amount	50 of 50	none
payer name	50 of 50	none

Document intake. Of the 50 letters (Table 10), muse-spark-1.3 read all eight fields correctly on 46 and seven of eight on the other 4. The 4 errors were one field and one glyph: the letter O read as a zero, in the monospace face the letter uses for codes. The reason code, the remark (copied verbatim in 50 of 50), both dates, the billed amount, the code system, and the payer's name were read correctly in every letter. The run cost $0.27.

Dictation. On the text-to-speech stand-in clip, the transcript came back in 5.5 seconds for $0.0014 with two doubled words the clip does not contain, and muse-spark-1.3 drafted and the gate passed 10 of 10 sentences, three of them quoting the dictated entry, doubled words included, for $0.028.

4 Discussion

4.1 Interpretation

The hypothesis held on this set. No model fabricated a quote in a sentence in seventeen runs; the two fabricated in evidence items were on drafts the model declined, and the gate refuted them. The spike-in shows why the absence is informative: every hard corruption injected into a verified sentence was caught, so a fabricated number, date, code, side, quote, or citation in a passing draft would have been found, unless the wrong value happened to appear in an admissible source (section 4.3). The cost of that guarantee is visible in the blocked column: nine drafts stopped for sentences that were true to the chart but cited one entry for a fact that lives in another, or stated a count the chart never states, and two empty warranted answers stopped for having nothing to verify. The design favors blocking a true derivation over passing an invented one, and the physician is told what to fix. Agreement stayed at or above 90 percent in eight of the nine kept runs; the ninth, Muse Spark Open with the live query, stands at 88 percent after v1.7 blocked a fifth true sentence, so the hypothesis held in eight runs and missed by one case in the ninth, and every one of that run's six disagreements is a blocked true sentence or the one false pass, not a fabrication.

The secondary question also answered itself. Given an index and two tools, the models read the chart the way a reviewer does: a third of the entries, almost all of the ones that matter, nothing at all for a denial the chart cannot answer, and nearly everything before declining a case where the evidence is absent.

4.2 The disagreements, read

muse-wonder-1.0 declined a wound-care case the key calls warranted. The criterion lists infection control and glycemic management among the components of standard wound care. The weekly notes document debridement, dressings, and offloading, and the chart holds a normal hemoglobin A1c and a clean wound elsewhere. The model read the criterion strictly and said the record did not meet it. That is a judgment a physician could share. It is not an invented fact.

The gate blocked a true count: "three antianginal agents," citing a note that lists metoprolol, amlodipine, and isosorbide. The count is correct and the gate blocked it, because the number three appears nowhere in the record. The gate cannot count. Whether to exempt simple counts, with the derivation recorded, is an open question.

Twice in the kept runs and three times in one repeat, a model set warranted to true on an absent case with no evidence and no sentences. The gate blocked every one. That is the fail-closed design doing its job, and it is also the clearest sign that declining is the harder half of the task.

More effort did not buy more agreement on this set (Table 9). For muse-spark-1.3 the added reasoning went into not committing: at high effort four answers set warranted to true and wrote nothing, three of them on cases the key labels unsupported, where the same model at low effort had declined with a reason. The gate blocked every one, so nothing false left, but an empty warranted answer is a worse answer than a decline, because it tells the physician nothing. For Muse Spark Open, medium effort gave the best run of any in this study, 50 of 50 with the case-023 draft declined, and high effort cost it two answers to the output cap at twice the latency. Effort low was fixed before the kept runs and stays the reported setting; these four files say that the open-weights model's disagreements at low were not a ceiling of the model, and that one run of fifty at one setting is too few to call medium its right setting.

The repair turn (Table 8) is the answer to the first two of these that costs nothing in the gate. Told which entry holds the number it stated, the model cited it, and six of the seven drafts blocked for a true derivation passed on the second turn at about a twentieth of a run's cost; the gate did not change and still cannot count. The seventh, the empty warranted answer on case-023, came back empty and stayed blocked, which is the behavior the design wants: a repair that argued a case the record does not support would be the repair eroding the decline, and in seven repairs none did. Seven is a small count, and the column that would show the erosion is in the table so a larger run can be read the same way. The stated reason changes nothing in the outcomes and everything in what a reader can see: with no thought text from the API, the one sentence the model writes before each read is the only account of why it opened what it opened, and in the three repair runs it never omitted one.

4.3 The referential limit

Case-023 is the most important result here, and it is on the public scoreboard rather than in a superseded run. The gate checks that every statement is in the record, not that the record satisfies the criterion. "Plan 5 to 6 days a week" answers the frequency half of a criterion and not the intensity half, and the gate passed it because the words are real. The LITMUS papers name this the referential limit: it is a referential warrant, and it verifies consistency, not correctness [11]. The spike-in puts a number on the same limit from the other side: 1,251 of 1,258 wrong values that happened to exist in an admissible source passed, and 2,382 changes with no hard token in them, a flipped negation or a swapped pronoun, passed (Table 6). The advisor caught case-023 and the gate cannot, which is the division of labor the design intends: the gate answers whether a statement is in the record, the advisor offers an opinion on whether the record satisfies the criterion, and the physician who signs is the check on the second question.

The post hoc run from the other chair (Table 7) says the same thing from the insurer's side. The gate held the plan's determination to the same record it holds the physician's appeal to, and it took no side: the one fabricated quote in a sentence in any full run, plan-exclusion language a model invented while arguing the insurer's case, was refuted, and a determination that misread a six-visit program as short of six months passed because its words were in the record. What the gate cannot see is the flag. In the probe, a model set the denial to stand and then wrote six sentences that each support the care; every sentence verified, and only a reader can see that the letter argues against its own conclusion. The standard applies to the insurer's algorithm as much as to this one, and so does its limit.

4.4 What the models found wrong with the test

Four of my grading keys were defective, and each was found by a model reading the chart more carefully than I had. A rehabilitation chart labeled warranted documented two physician visits in 14 days against a criterion of three a week. A sepsis chart labeled absent still carried a lactate and a blood pressure in a later note. A fusion chart labeled absent still mentioned the slip and the motion on the films. A carotid chart put surgery outside the 14-day window the criterion requires. Each was fixed by regenerating only the affected cases, and the ledger records every one. A benchmark written by one person has that person's blind spots, and the models were useful reviewers of it.

The same held for the code. After two cycles of the author's own multi-agent review, Meta's coding agent, Muse Code, reviewed the repository twice from a clone with no network and nothing writable: once on muse-spark-1.3 at effort high as a single agent, once on muse-wonder-1.0 at effort xhigh with the workspace

trusted. Both were asked to attack the gate with runnable inputs, and both found token classes the gate never had: a count written as "one" or "a", a date written with a month name, a side written R or L, a code whose letter or system word is wrong, a two-stopword quote that satisfies the verbatim bar, and a number recombined from a date or from a height and weight in the cited entry. None of these moved a stored verdict at the time, and none was among the 9,518 corruptions of the v1.5 spike-in, which is the point: the spike-in measures the corruption shapes its author thought of. The two reports, the prompt, and the complete traces of both runs are in the repository, unedited. The gate findings became six spike-in classes and gate v1.6 the same night, with the reproductions as tests, and the four classes the review left open closed as v1.7 the next evening; section 3 reports what each regate moved.

4.5 In context

The tools in the public record decide in about a second: Cigna's PxDx averaged 1.2 seconds per denial, and UnitedHealth's post-acute denial rate rose from 8.7 to 22.7 percent in four years [5, 6]. The response measured here takes four to sixteen seconds and a few cents, which is the first time the two sides of the exchange have run at comparable speed. The Inspector General's finding that Medicare Advantage plans applied clinical criteria not in Medicare's coverage rules [3] is the reason the design places the payer's criteria in force on the date of service beside the record rather than asking the model what the rules are. And the result that most changed the test, four grading keys corrected by models reading more carefully than their author, suggests a use for these models before any letter is drafted: as reviewers of the reviewer.

A filter circulated in October 2026 holds that an efficiency in United States healthcare nets to zero when both sides can buy the same model [17]. This design is built to pass it. The model is the part both sides can buy. The chart is held by one side only, and a denial is a claim about a policy while an appeal is a claim about the record. And the gate is not a model but a shared standard, published on its own, that lets the other side check what the record says rather than contest whose bot drafted better.

4.6 Limitations

The gate checks consistency, not correctness. It verifies that the quoted words are in the cited entry and that every number, date, code, and side in the sentence is sourced. It does not verify that those words satisfy the criterion. A sentence that carries no hard token is verified by any quote that exists in the entry it cites. Case-023 is the measured instance, and the physician's signature is the design's answer to it.

The gate cannot derive. A count read off a list, a duration computed from two dates, or a side inferred from an entry that names it elsewhere is blocked unless the token appears verbatim. From v1.6 this includes a count written as "one" where the entry says "within the hour", and from v1.7 a count written as "a week" or "a year" where the entry says "this week" or "15 months": eleven true sentences across the runs are blocked for it, four of them in the kept runs.

Token classes remain outside the gate: a count derived from a list, a Roman numeral with no grading word before it, a side-number pair cut across a conjunction or punctuation the clause rule does not recognize, and any frequency or discourse idiom the fixed mask lists do not name; each is a candidate spike-in class.

A sentence is not checked against its own quote. The gate requires the quote to be in the entry and every hard token in the sentence to be sourced somewhere admissible; it does not require the sentence's numbers to agree with the quote beside them. The collision class of the spike-in measures exactly this, and about a fifth of those sentences contradict their own quote and pass. Checking the sentence against the quoted span is the next rule to add, with the derivation cases above as the cost to weigh.

Soft words are not checked. A model once wrote "he" for a female patient and nothing tripped the gate. Sex, age, and alias are case facts and could be checked next; the spike-in gives the size of the blind spot.

One label in the key was corrected after the first deposit. Case-023 was labeled absent, meaning no entry addresses the intensity criterion, yet its screening note says "three disciplines at this intensity" and a progress note "the first full therapy day", lines it shares with its near-miss sibling case-034. By the key's own definitions the chart raises the criterion without meeting it, so on October 30 the label was changed to near-miss in the generator, with the reason in the case's note, and the stored result files were updated to carry it; the

chart text and every verdict and hash are unchanged, the archived version carries the old label, and the one false pass still sits on this case.

The counts are small, and the author graded his own test. Fifty synthetic cases, one seed. The charts, the criteria, and the key were written by one person, which makes the test easier than a real chart against a real policy and leaves blind spots the models found four of. Six of the nine kept rows are single runs.

The data are synthetic. No result here says anything about performance on real records, and that measurement cannot be made in public.

The denials are post-service claims. The prior-authorization refusal before care, which the statistics in section 1.1 describe, is the same document answered in a different process, and it was not tested.

Cost and latency are the API's. Muse Spark Open reported no reasoning tokens on the Baseten path, so its cost may be understated; latency is wall-clock from one client and not a practice's network.

4.7 What should come next

A prospective study inside a practice, under its own governance and not in public, with the endpoints that matter: physician minutes per appeal, the share of drafts signed without edit, and the overturn rate against the practice's own history. That practice exists; the author's own practice runs on the pre-existing systems named in the Declarations, so the study waits on governance, not on building. Then prior-authorization cases and the peer-to-peer; sex, age, and alias as checked facts, and a recorded derivation for simple counts; a second seed and an independent author for the key, with inter-rater agreement reported; real policy text from a coverage index in place of synthetic criteria; and the gate and spike-in harness used on other drafting systems, since the standard applies to an insurer's algorithm as much as to this one.

5 Conclusion

A physician can now answer a denial at the speed and cost at which it was issued, with every statement held to their own record by a check that has not let a fabrication through in 8,602 tries. The check verifies presence, not sufficiency. The machine prepares the instrument; it cannot issue it. The physician's signature is what makes it a writ.

Acknowledgements

Meta and Cerebral Valley for the event; Baseten for hosting the open-weights model; Vercel for hosting the replay. The three models under test found four defects in the grading key, and the key is better for it.

Declarations

Competing interests. I am the author of the LITMUS papers and of three pre-existing systems of mine, an electronic health record, a denial and appeal engine, and the coverage index queried in section 2.5, that this work does not include but that a production version would run on, and I have a commercial interest in them. I am a practicing surgeon whose claims are subject to the denials described here. The models under test, muse-spark-1.3, muse-wonder-1.0, and Muse Spark Open, are Meta's; the work was done for a Meta-sponsored event, and event credits on Baseten paid for part of the runs reported; the cost of every run is in its result file. No payer, insurer, or model vendor had any role in the design, the cases, the key, or the reporting.

Ethics. All data are synthetic. No human subjects, no patient data, no institutional review. The generator emits no names, dates of birth, record numbers, member numbers, or claim numbers, and a test checks the committed files against the generator.

Funding. None beyond the event credits named above.

Data and code availability. The set, the gate, the harness, every result file, the mutation reports, the advisor output, the letters, the review traces, and the viewer are in the repository under Apache-2.0; the written content is CC BY 4.0. The repository is deposited at Zenodo under a concept DOI that resolves to the newest version and lists all versions: v1.0.0 (the kept runs, repeats, spike-in, and payer-side runs), v1.1.0 (the post hoc work of section 2.8), and v1.2.0 (gate v1.7 and the regate, the npm package, the phone viewer) [14]. This manuscript describes v1.2.0 and is itself deposited at Zenodo as a preprint, doi:10.5281/zenodo.24000088. The live replay is at demo.hemeehr.com [15], and the gate is on npm as litmus-gate [16].

References

1. KFF. Nearly 50 million prior authorization requests were sent to Medicare Advantage insurers in 2023. January 28, 2025. https://www.kff.org/medicare/issue-brief/nearly-50-million-prior-authorization-requests-were-sent-to-medicare-advantage-insurers-in-2023/. The 81.7 percent overturn rate is for appeals decided in 2023, as summarized by AHA News, January 29, 2025. KFF's page now also carries the 2024 figures, which are larger: 52.8 million requests, 4.1 million denied (7.7 percent), 80.7 percent of appeals overturned.
2. American Medical Association. AMA survey: prior authorization reform pledge falls short with physicians. May 2026. https://www.ama-assn.org/press-center/ama-press-releases/ama-survey-prior-authorization-reform-pledge-falls-short-physicians
3. Office of Inspector General, U.S. Department of Health and Human Services. Some Medicare Advantage Organization denials of prior authorization requests raise concerns about beneficiary access to medically necessary care. OEI-09-18-00260, April 27, 2022. https://oig.hhs.gov/reports/all/2022/some-medicare-advantage-organization-denials-of-prior-authorization-requests-raise-concerns-about-beneficiary-access-to-medically-necessary-care/
4. DistilINFO. Court orders UnitedHealth to disclose AI denial algorithm. March 12, 2026. https://distilinfo.com/2026/03/12/court-orders-unitedhealth-to-disclose-ai-denial-algorithm/
5. U.S. Senate Permanent Subcommittee on Investigations. Majority staff report on Medicare Advantage. October 17, 2024. https://www.hsgac.senate.gov/wp-content/uploads/2024.10.17-PSI-Majority-Staff-Report-on-Medicare-Advantage.pdf. Summary: Healthcare Dive, https://www.healthcaredive.com/news/medicare-advantage-AI-denials-cvs-humana-unitedhealthcare-senate-report/730383/
6. Healthcare Dive. Cigna sued over PxDx claim denials. July 2023. https://www.healthcaredive.com/news/cigna-lawsuit-algorithm-claims-denials-california/
7. The Wall Street Journal. Insurers pocketed $50 billion from Medicare for diseases no doctor treated. July 2024; and The one-hour nurse visits that let insurers collect $15 billion from Medicare. August 2024. Summaries: Rise Health, July 9, 2024, https://www.risehealth.org/insights-articles/article/wsj-investigation-medicare-advantage-plans-were-paid-billions-for-diseases-patients-never-had/; KFF Health News, https://kffhealthnews.org/morning-breakout/insurers-bilked-50b-from-medicare-for-dubious-diagnoses-review-finds/
8. Medical Economics. UnitedHealth Group under DOJ investigation over Medicare billing practices. July 24, 2025. https://www.medicaleconomics.com/view/unitedhealth-group-under-doj-investigation-over-medicare-billing-practices
9. Medical Economics. Lawmakers on the WISeR model. May 2026. https://www.medicaleconomics.com/view/medicare-wiser-prior-authorizations-needed-congress-prior-authorization-lawmakers-say
10. Centers for Medicare and Medicaid Services. WISeR model. https://www.cms.gov/priorities/innovation/innovation-models/wiser
11. Rosing JH. Litmus: The Admissible Set. Zenodo, July 6, 2026. doi:10.5281/zenodo.21227402
12. Rosing JH. Litmus: The Admissible Set (System Architecture). Zenodo, July 6, 2026. doi:10.5281/zenodo.21227855
13. Meta Global AI Developer Hackathon, hosted by Meta and Cerebral Valley, October 26 to November 4, 2026. https://cerebralvalley.ai/e/meta-ai-developer-hackathon
14. Rosing JH. Your procedure is denied: fail-closed verification of model-drafted claim-denial appeals against the clinical record. Software and data, Zenodo, 2026. Concept doi:10.5281/zenodo.23794797;

v1.2.0 doi:10.5281/zenodo.23935951. Source: https://github.com/jamesrosing/your-procedure-is-denied
15. The live replay. https://demo.hemeehr.com
16. litmus-gate, version 0.3.0. npm, November 2, 2026. https://www.npmjs.com/package/litmus-gate
17. Cuban M. On why operational AI struggles in United States healthcare, as circulated on LinkedIn, October 2026: there is no price discovery, so savings never become lower prices; prior-authorization and billing bots fight bots on the other side running the same models; and if both sides can buy the same model, it nets out to zero.

Supplementary material

S1. The figures as replay frames. The replay at demo.hemeehr.com addresses any case of any kept run by deep link; step=end lands on the verdict and the arrow keys walk the trace from there. The following frames are the figures of this paper beyond Figures 1 and 2.

- S1.1. A warranted case drafted and verified: case-001, muse-spark-1.3 with the live query, at the verdict. Eight sentences, each stamped against the entry it cites; the payer's reason above the criteria it is answered by. https://demo.hemeehr.com/#file=meta-muse-spark-1.3-low-agentic-chart-2026-10-28T06-48-41-422Z.json&case=case-001&step=end
- S1.2. The gate blocking a true sentence: case-038, Muse Spark Open with the live query, at the verdict. The count of three antianginal agents, true to the chart, cited to a note that lists the agents without the number; the gate names the token, returns needs-citation, and names the entry that holds it. https://demo.hemeehr.com/#file=baseten-meta-models_muse-spark-open-low-agentic-chart-2026-10-28T07-13-57-225Z.json&case=case-038&step=end
- S1.3. A decline: case-044, muse-wonder-1.0 with the live query, at the verdict. All 22 entries read before the model declined, with its reason. https://demo.hemeehr.com/#file=meta-muse-wonder-1.0-low-agentic-chart-2026-10-28T06-51-47-640Z.json&case=case-044&step=end
- S1.4. The referential limit: case-023, Muse Spark Open with the live query, at the verdict. The quote "plan 5 to 6 days a week" beside the intensity criterion it does not meet, the gate's pass, and the advisor's dissent. https://demo.hemeehr.com/#file=baseten-meta-models_muse-spark-open-low-agentic-chart-2026-10-28T07-13-57-225Z.json&case=case-023&step=end
- S1.5. The live policy query: case-001, muse-spark-1.3, step 1. The payer, the authority asked, the index, and the answer, labeled as a query against a prior-work data source. https://demo.hemeehr.com/#file=meta-muse-spark-1.3-low-agentic-chart-2026-10-28T06-48-41-422Z.json&case=case-001&step=1
- S1.6. The gate in the browser: case-001 at the verdict, then "test the gate" on the fourth sentence. Change 110 feet to 150 feet and the gate refutes the sentence, naming the token; change it to 300 feet, a figure that lives in a later entry, and it returns needs-citation, naming that entry. Nothing is saved.
- S1.7. The instrument: case-001 at the verdict, then "the writ" on the pass line. Eight sentences, eight footnotes to the record, the gate's hash, and the blank signature line under the band: draft, unsigned, not sent.
- S1.8. The other chair, one chart, two verified conclusions: case-036, bariatric surgery, the physician's appeal (muse-spark-1.3 with the live query, pass, eight sentences) and the plan's determination (muse-spark-1.3 arguing the insurer's side, pass, six sentences), the fourth reading "supervised weight management visit 6 of 6" as short of six consecutive months. https://demo.hemeehr.com/#file=meta-muse-spark-1.3-low-payer-2026-10-30T06-12-28-712Z.json&case=case-036&step=end
- S1.9. The repair turn: case-038, muse-spark-1.3 with one repair turn, at the first verdict (step 21), blocked, sentence 1 needs-citation for the count of three antianginal agents, the gate naming note-003; one step on, the repair turn with that line sent back verbatim; at the verdict, pass after the repair turn, sentence 1 re-stamped verified with its first stamp kept above it. https://demo.hemeehr.com/#file=meta-muse-spark-1.3-low-agentic-chart-repair1-2026-10-31T06-25-01-670Z.json&case=case-038&step=end
- S1.10. The stated reason: case-001 on the same run, step 2, the first entry opened and, above it, the sentence the model wrote before opening it. https://demo.hemeehr.com/#file=meta-muse-spark-1.3-low-agentic-chart-repair1-2026-10-31T06-25-01-670Z.json&case=case-001&step=2

- S1.11. Three models, one case: case-001 on the Compare tab, the three kept live-query results side by side. https://demo.hemeehr.com/#file=meta-muse-spark-1.3-low-agentic-chart-2026-10-28T06-48-41-422Z.json&case=case-001&view=compare
- S1.12. The denial as it arrives: case-001 as a remittance letter image, the image sent to the intake step; its eight fields were read back correctly except the code, KDOB1 read as KD0B1. In the repository under build/synthetic/letters/case-001.png.

S2. Commands. In the repository's build directory: npm test runs the 282 tests; npm run synthetic regenerates the fifty cases; npm run harness runs a model over the set; npm run mutate reruns the spike-in; npm run regate re-gates every stored result under the current gate; npm run log regenerates the build log and the tables; npm run gate -- check draft.json case.json returns a verdict with exit code 0, 1, or 2 for pass, blocked, or declined; npx -y litmus-gate mcp serves the gate to any agent host over stdio.